



OPEN Leukocyte telomere attrition following radiotherapy in prostate cancer: a prospective study

Tanja Langsenlehner¹, Katarzyna Paal¹, Eva- Maria Thurner¹, Selina Genser², Renate Sternat², Bettina Stranz¹ & Wilfried Renner²✉

Telomeres are protective protein-bound DNA repeat structures at the end of chromosomes, which play a critical role in maintaining chromosomal stability. With each somatic-cell division, telomeres progressively shorten, making telomere length a potential biomarker of biological aging. Ionizing radiation may accelerate telomere attrition, thereby promoting aging-related changes. In the present study, we analyzed the influence of radiotherapy on leukocyte telomere length in prostate cancer patients. A total of 314 patients treated with curative radiotherapy for prostate cancer were included in the present prospective study. Leukocyte relative telomere length (RTL) was measured by qPCR in peripheral blood samples collected before radiotherapy, at the end of radiotherapy, and at 3 and 15 months post-radiotherapy. Mean RTL values were 0.65 ± 0.34 at baseline, 0.62 ± 0.31 at the end of radiotherapy, 0.67 ± 0.43 , and 0.55 ± 0.26 at the first and at the second follow-up, respectively. Paired-Samples T-Test comparisons showed a significant reduction in RTL at 15 months post-radiotherapy compared to baseline ($p < 0.001$), end of radiotherapy ($p = 0.001$), and 3-month follow-up examination ($p < 0.001$). In our cohort, we observed a significant shortening of telomeres after radiotherapy indicating a potential contribution to accelerated cellular aging.

Keywords Prostate cancer, Radiotherapy, Aging, Cellular senescence, Telomere length

Cancer treatments can cause the early onset of age-related chronic conditions such as frailty, sarcopenia, heart dysfunction, and cognitive decline. While age-related health changes are observed in the majority of cancer survivors, the understanding of the aging effects of cancer treatments and the factors contributing to varying responses to therapy is still limited.

Aging is a complex, time-related process characterized by the accumulation of molecular and cellular damage and loss of physiological integrity. An acceleration of aging has been noted following chemotherapy and radiation, however, the mechanisms behind this accelerated aging in survivors remain unclear. Evidence from preclinical models shows that radiation and cytotoxic anticancer therapies cause physiological changes consistent with several molecular and cellular hallmarks of aging^{1,2} such as genomic instability, telomere attrition, epigenetic alterations, mitochondrial dysfunction, loss of proteostasis, chronic low-grade inflammation, and cellular senescence^{3,4}.

Cellular senescence represents a stress-induced, durable cell-cycle arrest of previously replication-competent cells and can be triggered by a variety of cell-intrinsic and extrinsic stimuli. Although there are no senescence-specific biomarkers, a few markers have been suggested to identify senescent cells with some confidence.

There is growing evidence indicating a significant correlation between telomere length and cellular senescence. Telomeres are protein-bound DNA repeat structures at the end of chromosomes that play a critical role in maintaining chromosomal stability⁵. During somatic-cell replication, telomere length progressively shortens because of the inability of DNA polymerase to fully replicate the 3' end of the DNA strand. Once a critically short telomere length is reached, the cell is triggered to enter replicative senescence, which subsequently leads to cell death⁶. In cross-sectional population studies, mean telomere length is inversely associated with age declining by between 20–40 base pairs per year⁷. Within an individual, mean leukocyte telomere length and telomere length in other tissues are highly correlated.

Besides aging, many different endogenous and environmental factors, such as oxidative stress, an unhealthy lifestyle and genotoxic stress, can modify telomere length. Among young adult survivors of childhood acute

¹Department of Therapeutic Radiology and Oncology, Comprehensive Cancer Center, Medical University of Graz, 8036 Graz, Austria. ²Clinical Institute of Medical and Chemical Laboratory Diagnostics, Medical University of Graz, 8036 Graz, Austria. ✉email: wilfried.renner@medunigraz.at

lymphoblastic leukemia, shorter leukocyte telomere length has been detected when compared with age-matched controls⁴. Similar results have been obtained in long-term survivors of neuroblastoma suggesting premature senescence in leukocytes of the survivors⁵. In addition, a variety of chemotherapeutic agents have been implicated in the pathogenesis of senescence and acquired telomeropathies, associated with frailty phenotypes and aging-associated diseases⁹.

However, data on the aging-related consequences of radiotherapy are sparse. In the present prospective study, we investigated the influence of radiotherapy on cellular senescence measured with leukocyte relative telomere length (RTL). End of radiotherapy: Captures the acute biological impact of ionizing radiation on circulating leukocytes, when radiation-induced DNA damage, oxidative stress, cell turnover, and potential telomere shortening or lengthening mechanisms are most pronounced.

3 months post-radiotherapy: Represents the early recovery phase, allowing assessment of short-term telomere dynamics after treatment cessation, including hematopoietic repopulation, resolution of acute inflammation, and early adaptive or compensatory telomere maintenance processes.

15 months post-radiotherapy: Reflects long-term or persistent effects, enabling evaluation of delayed or chronic radiation-associated telomere alterations, long-term genomic stability, and potential links to late radiation toxicity or aging-related processes.

Materials and methods

This prospective observational study was performed including 314 prostate cancer patients from the “RADIOtherapy: Side effects of Treatment and Aging Related conditions” (RADIOSTAR) study. RADIOSTAR is a prospective longitudinal study aimed to investigate (1) the influence of radiotherapy on aging-related processes, (2) the association between aging related biomarkers and clinical conditions, and (3) the relationship of age-related clinical and molecular changes with radiation-induced toxicity, quality of life, and major outcome measures.

Participants of RADIOSTAR were male patients aged ≥ 65 years with sporadic, histologically confirmed prostate cancers treated with curative therapy at the Department of Therapeutic Radiology and Oncology, Medical University of Graz.

Data on tumor and treatment characteristics, including prostate-specific antigen (PSA) level at diagnosis, pre-irradiation PSA level, tumor-node-metastasis (TNM) stage, Gleason score, and details on androgen deprivation therapy (ADT), were collected from all participants.

Image-guided radiotherapy with high-energy photons was typically administered using volumetric intensity-modulated arc therapy (VMAT) techniques to cover the prostate and seminal vesicles. The total dose, prescribed to the International Commission on Radiation Units and Measurements (ICRU) reference point, ranged from 70 to 78 Gy in 2 Gy fractions, depending on the risk category. Additional pelvic node irradiation with a total dose of 50.4 Gy in 1.8 Gy fractions was performed in 7 patients. Hypofractionated radiotherapy with a total dose of 60 Gy (3 Gy per fraction) was administered in 13 patients. Treatment was performed daily, 5 days/week.

EDTA-blood was drawn at baseline (within 2 weeks before radiotherapy), at the end of radiotherapy, and 3 and 15 months post-radiotherapy. Time points were chosen for the detection of acute biological impact of ionizing radiation on circulating leukocytes (end of radiotherapy), assessment of short-term telomere dynamics after treatment cessation (3 months post-radiotherapy) and analysis of long-term or persistent effects (15 months post-radio-therapy).

Genomic DNA was prepared from whole blood using a MagNA Pure 24 System (Roche Austria GmbH, Vienna, Austria). RTL was measured by a qPCR assay developed by Cawthon¹⁰ with minor modifications. The assay quantified the ratio of average telomere length to a single-gene copy. The method used for measurement of RTL provides only relative length and results cannot be converted to absolute base-pairs. All qPCR analyses were performed in 96-well plates on a CFX96 Real-Time PCR Detection System (Bio-Rad Laboratories, Vienna Austria). Total reaction volume was 25 μ l. Thermal cycling profile was 10 min at 95 °C followed by 40 cycles of 15 s at 95 °C, 30 s at 54 °C and 30 s at 72 °C with fluorescence data collection. Each run included a standard curve made by dilutions of available HeLa-DNA (New England Biolabs, Frankfurt, Germany) to determine quantity of targeted templates in each sample relative to the HeLa-DNA. RTL was calculated as the ratio of telomere quantity to single copy gene quantity. Mean intra-assay coefficient of variability (CV) was 8.6% and plate-to-plate CV was 12.6%.

RTL was measured only in within the PROCAGENE cohort. No group of healthy control subjects was analyzed in the present investigation.

Ethical considerations

The study complies with the Declaration of Helsinki and has been conducted in accordance with national law. The protocol was approved by the local Ethics Committee (approval number: EK 31–437 ex 18/19). Data are stored electronically with strict adherence to data protection regulations, including the Austrian Gene Technology Act, the Data Protection Regulation, and the guidelines of the Ethics Committee of the Medical University of Graz. Written informed consent was obtained from all participants.

Statistical analyses

The primary endpoint was the occurrence of changes in RTL during the observation period. Descriptive statistics were used to summarize clinical and treatment-related parameters. Changes in telomere length were analyzed using paired-samples T-Test. Linear regression models were applied to assess the impact of patient and treatment factors on telomere length changes. P-values ≤ 0.05 were considered statistically significant. Statistical analyses were performed using SPSS 28.0 for Windows (SPSS Inc., 1989).

Results

Patient and treatment characteristics are presented in Table 1. Median age at start of radiotherapy was 73 years (mean 73.55 ± 5.58). Follow-up investigations were performed 3 and 15 months after completion of radiotherapy. A total of 261 patients underwent the first follow-up investigation, the second follow-up investigation was performed in 244 patients.

Mean RTL values were 0.65 ± 0.34 kb at baseline, 0.62 ± 0.31 kb at the end of radiotherapy, 0.67 ± 0.43 kb at the first follow-up, and 0.55 ± 0.26 kb at the second follow-up, respectively (Fig. 1).

Pairwise analyses revealed a significant decrease in RTL 15 months after radiotherapy compared to baseline ($p < 0.001$), as well as compared to measurements performed at the end of radiotherapy and 3 months post-radiotherapy (both $p < 0.001$). However, no significant differences were observed between baseline and the measurements at the end of radiotherapy or 3 months post-radiotherapy ($p = 0.60$ and $p = 0.82$, respectively). Additionally, no significant change in RTL was detected between the measurements at the end of radiotherapy and 3 months post-radiotherapy ($p = 0.50$). Changes in telomere length are displayed in Table 2.

Seven patients received additional pelvic node irradiation and thirteen patients hypofractionated radiotherapy. RTL changes did not reach statistical significance in these small subgroups.

Additionally, we analyzed the impact of clinical and treatment-related factors on RTL changes identified as significant in the non-parametric tests (Table 3). A significant association was observed between age and RTL changes 15 months after radiotherapy compared to baseline ($p = 0.009$). Additionally, hypertension was significantly associated with RTL changes between 15 and 3 months post-radiotherapy ($p = 0.045$). A trend was also noted for the relationship between nodal irradiation and RTL changes from baseline to 15 months post-radiotherapy ($p = 0.070$), as well as for an association between smoking and RTL changes between the end of radiotherapy and 15 months post-treatment ($p = 0.091$).

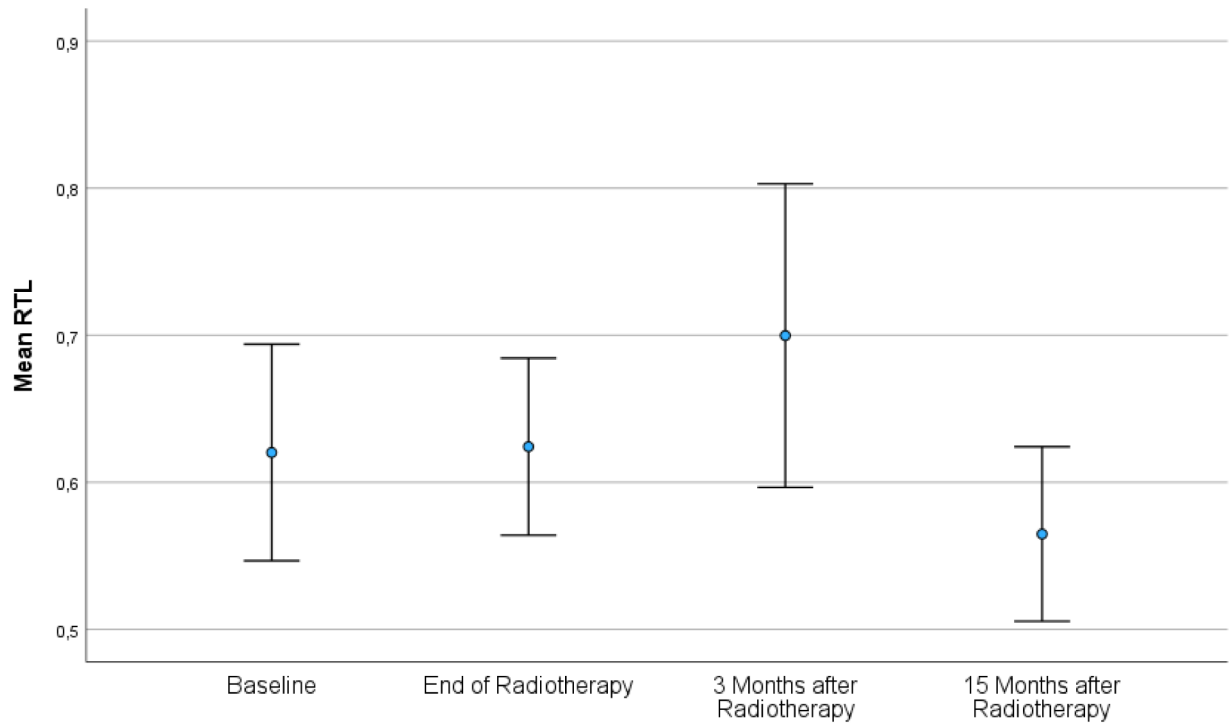
Discussion

In the present study, we investigated the effects of prostate cancer radiotherapy on telomere length and detected a significant telomere attrition 15 months after radiotherapy when compared to the assessments at baseline, at the end of RT, and 3 months post-RT. Our results indicate that exposure to ionizing radiation may accelerate telomere shortening, potentially leading to genomic instability and increased susceptibility to cellular senescence or apoptosis.

One of the key mechanisms through which radiotherapy may induce telomere shortening is oxidative stress. Ionizing radiation generates reactive oxygen species (ROS) that cause oxidative damage to DNA, including telomeric regions, which are particularly vulnerable due to their high guanine content¹¹. Unlike double-strand breaks in the coding regions of the genome, damage to telomeres is less efficiently repaired, exacerbating telomere

Parameter		N (%)
Age	< 70 years	68 (21.7%)
	≥ 70 years	246 (78.3%)
Smoking status	Current	13 (4.2%)
	Not current	300 (95.8%)
Hypertension	Yes	177 (56.5%)
	No	136 (43.5%)
Diabetes	Yes	56 (17.9%)
	No	257 (82.1%)
Comorbidities	CCI 0	187 (59.4%)
	CCI 1–2	80 (28.2%)
	CCI ≥ 3	17 (6.0%)
Risk group	Low risk	28 (8.9%)
	Intermediate risk	139 (44.3%)
	High risk	147 (46.8%)
Androgen deprivation	None	18 (5.7%)
	Neoadjuvant	104 (33.1%)
	Neoadjuvant and adjuvant	192 (61.0%)
Fractionation	Conventional	301 (95.9%)
fractionation	Hypofractionation	13 (4.1%)
Total RT dose	≤ 76 Gy	121 (38.5%)
	76 Gy	193 (61.5%)
Nodal irradiation	Yes	7 (2.2%)
	No	306 (97.8%)

Table 1. Patient and treatment characteristics. Gy Gray, RT radiation therapy, CCI Charlson Comorbidity Index.



Abbreviations: RTL – relative telomere length

Fig. 1. Mean RTL values at baseline, at the end of radiotherapy, at the first and second follow-up examination. RTL significantly decreased 15 months after radiotherapy compared to baseline, end-of-therapy, and 3-month post-therapy measurements (all $p < 0.001$). No significant differences were found between baseline and end-of-therapy or 3-month post-therapy measurements ($p = 0.60$ and $p = 0.82$, respectively), nor between end-of-therapy and 3-month post-therapy measurements ($p = 0.50$).

	Relative telomere length	
	Median difference (mean ± SD)	P-value
Baseline – end of RT	0.015 (0.019 ± 0.436)	0.601
Baseline – 3 months post-RT	-0.039 (-0.027 ± 0.444)	0.816
Baseline – 15 months post-RT	0.088 (0.125 ± 0.367)	<0.001
End of RT – 3 months post-RT	0.012 (-0.059 ± 0.442)	0.504
End of RT – 15 months post-RT	0.116 (0.112 ± 0.336)	<0.001
3 months post-RT – 15 months post-RT	0.117 (0.228 ± 0.480)	<0.001

Table 2. Relative telomere length before and at the end of radiotherapy as well as 3 and 5 months after radiotherapy. RT radiation therapy, SD standard deviation. Bold values indicate significant p-values.

loss with each subsequent cell division^{3,12}. In addition to ROS-mediated damage, radiation-induced disruption of telomerase activity has been observed, further impairing the capacity of cells to preserve telomere length¹³. Telomeres, the protective caps at the ends of chromosomes, play a critical role in maintaining genomic stability. Telomere shortening is a key feature of cellular aging, linked to age-related diseases, cancer, senescence, and stem cell exhaustion. Shorter age- and sex- adjusted mean telomere length has been associated with various aging

Parameter	Baseline – 15 months post-RT			End of RT – 15 months post- RT			3 months – 15 months post- RT		
	Coefficient	Standardized coefficient	p-value	Coefficient	Standardized coefficient	p-value	Coefficient	Standardized coefficient	p-value
Age (< 70 vs. ≥ 70)	0.196	0.236	0.009	-0.005	-0.006	0.953	0.106	0.095	0.350
Smoking status (Current vs. not current)	-0.162	-0.110	0.230	0.210	0.171	0.091	-0.184	-0.099	0.331
Hypertension (Yes vs. no)	-0.013	-0.018	0.844	-0.055	-0.082	0.422	-0.193	-0.202	0.045
Diabetes (Yes vs. no)	-0.070	-0.077	0.404	0.108	0.124	0.221	-0.094	-0.081	0.428
Comorbidities (CCI 0 vs. 1–2 vs. ≥ 3)	0.020	0.033	0.725	0.067	0.124	0.232	-0.127	-0.160	0.124
Risk group (Low vs. intermediate vs high risk)	-0.024	-0.042	0.651	-0.071	-0.140	0.167	-0.060	-0.078	0.442
Androgen deprivation (None vs. neoadjuvant vs neo- and adjuvant)	-0.009	-0.015	0.867	-0.074	-0.142	0.161	-0.078	-0.101	0.319
Fractionation (Conventional vs hypofractionation)	-0.067	-0.040	0.664	0.086	0.066	0.517	0.086	0.066	0.517
Total RT dose [Gy] (≤ 76 vs. > 76)	0.002	0.002	0.981	-0.085	-0.126	0.215	0.003	0.031	0.759
Nodal irradiation (Yes vs. no)	0.389	0.166	0.070	-0.295	-0.151	0.135	0.433	0.155	0.125

Table 3. Influence of clinical and treatment parameters on changes in telomere length. Gy Gray, RT radiation therapy, CCI Charlson comorbidity index. Bold values indicate significant p-values.

related- diseases and morbidity, including coronary artery disease, diabetes mellitus, hypertension, Alzheimer's disease, and cancer^{14,15} and has also been linked with functional status, cognitive function or mortality^{16,17}.

Telomere shortening can be accelerated by various endogenous and environmental factors, such as psychological and physiological stress, unhealthy lifestyle, and genotoxic stress, including cigarette smoking, obesity, and high homocysteine blood levels¹⁸. Additionally, a correlation between cancer therapy and telomere shortening has also been reported^{19,9}.

In a previous study, we demonstrated that prostate cancer radiotherapy leads to significant changes in inflammatory markers associated with systemic inflammation²⁰. This systemic inflammatory response may accelerate leukocyte telomere attrition by promoting rapid cell turnover and/or by inducing the release of reactive oxygen species, which can directly damage telomeric DNA through oxidative stress or inhibition of telomerase activity^{21,22}.

In contrast, the direct damage of radiotherapy to the hematopoietic bone marrow is primarily reflected by changes of the leukocyte count. Although beyond the scope of the present study, we observed in our cohort that changes in leukocyte counts exhibited different temporal dynamics compared to changes in telomere length. A significant reduction in leukocyte count was observed between the start and end of radiotherapy, which was significantly associated with the bone marrow volume receiving a dose of at least 20 Gy ($p = 0.003$). This decrease was followed by a significant increase 3 months after therapy.

We therefore assume that the reduction in telomere length is more likely a consequence of the inflammatory response to radiotherapy than of damage to the hematopoietic bone marrow. It can also be hypothesized that, due to the larger volume of irradiation, the inflammatory response may be more pronounced after additional lymph node irradiation than after prostate-only irradiation.

Mean leukocyte telomere length is an important measure of cellular aging, with shorter telomeres typically linked to aging and various diseases. Thus, measuring mean leukocyte telomere length is a valuable tool for understanding biological age and health risks.

The shortening of telomere length may also be associated with faster aging and the earlier onset of age-associated changes observed in patients after cancer and its treatment, beyond treatment-related side effects.

Both in vitro and in vivo studies have demonstrated that irradiation triggers senescence in human cells²⁷. In young adult survivors of childhood ALL, leukocyte telomere length (LTL) has been found to be shorter compared to age-matched controls, with the most pronounced differences observed in those who received cranial radiotherapy⁴. Furthermore, asymptomatic cancer survivors have been found to exhibit a biological profile marked by chronic inflammation and telomere shortening, indicating early activation of cellular pathways associated with accelerated aging^{8,9}.

In contrast, some studies have reported telomere lengthening following exposure to exogenous stressors. For example, a large observational study of workers involved in the cleanup of the Chernobyl nuclear disaster site in 1986 found significantly longer telomeres in those with higher radiation exposure²³. However, the cohort described in the Chernobyl literature is fundamentally different from the cohort in our study, and therefore, the findings are limited in their comparability to ours.

Similarly, a cross-sectional study of ovarian cancer patients treated with surgery and cisplatin-based chemotherapy observed increased telomere length after therapy compared to baseline²⁴. Although not statistically significant, a slight increase in telomere length was also observed at the end of treatment compared to baseline in our study. This phenomenon may be explained by a transient increased telomerase activation.

Typically, the average telomere shortening in individuals over 60 years of age is estimated at 30 to 60 base pairs (bp) per year²⁵, which is roughly 0.5% to 1.0% of a typical 6 kb telomere in individuals of that age. In our study, 15 months after the completion of therapy, a significant reduction of 15% compared with baseline was observed, which was more pronounced than expected at this time point, suggesting that prostate cancer radiotherapy may substantially accelerate telomere attrition.

Telomere shortening may contribute to the long-term side effects of radiotherapy, including tissue fibrosis, premature aging, and secondary malignancies^{26,27}. This is particularly concerning for pediatric and young adult cancer survivors, as accelerated telomere attrition may predispose them to early-onset aging-related diseases. It is also a significant concern for individuals at advanced age, who may be especially vulnerable to the effects of accelerated aging.

To the best of our knowledge, this study represents the first analyzing the influence of small-volume prostate cancer radiation therapy on biochemical and molecular aging-related parameters. To date, no previous studies on aging have been performed comprising a comparable number of participants who obtained the same treatment regimen. Major strengths of our study include its prospective design as well as the inclusion of a large and homogeneous patient cohort with respect to age, cancer type, and prescribed treatment resulting in a high level of scientific significance. The selected observation period following the completion of radiotherapy may also enable a meaningful assessment of treatment-induced aging-related changes, and allow for detecting significant molecular, biochemical, and clinical alterations beyond those attributable to natural aging alone.

The limitation of our study is data collection from only one patients' collective without control group, which could enable a comparison with the dynamics of natural aging process. However, constructing a control group without any malignancy would introduce substantial selection bias. Moreover, potential confounders such as chronic psychological stress, which could not be fully accounted for, may have impacted our findings.

Therefore, further studies are necessary before definitive conclusions on the consequences of curative radiation therapy on the manifestations of aging can be drawn. Longitudinal studies monitoring telomere length in radiotherapy patients could offer further insights into the long-term effects of radiation exposure and potential strategies for improving patient outcomes. Future research should also focus on elucidating individual variability in radiation-induced telomere shortening. In addition, investigating potential interventions, such as antioxidant therapy or telomere-protective agents, may offer an opportunity for reducing radiation-associated genomic damage.

In conclusion, our data indicate that radiotherapy may accelerate telomere attrition, potentially leading to increased susceptibility to cellular senescence that could contribute to the development, acceleration or worsening of age-related alterations and conditions. Nevertheless, further research is necessary before definitive conclusions can be made regarding the impact of curative radiation therapy on aging-related manifestations.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request. Data are located in controlled access data storage at the Medical University of Graz.

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Author contributions

TL and WR wrote the main manuscript text. WR, TL, SG and RS analyzed data. TL prepared Fig. 1. KP, EMT and BS collected data. All authors reviewed the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to W.R.

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